

YouTube Trending Video Analytics Report

Introduction:

YouTube is one of the largest video platforms in the world where millions of videos are uploaded every day. Understanding trending videos helps to analyze user behavior, content popularity, and engagement patterns.

This project focuses on analyzing YouTube trending videos across multiple countries to identify key trends and insights using real-world data.

Abstract:

This project analyzes a real-world YouTube trending dataset to understand how different types of videos perform across countries. The data was cleaned, processed, and visualized using analytical tools. An interactive dashboard was created to show insights such as total views, likes, engagement, and trending duration. The project helps in understanding data visualization, dashboard design, and real-world data handling.

Tools Used:

The following tools and technologies were used in this project:

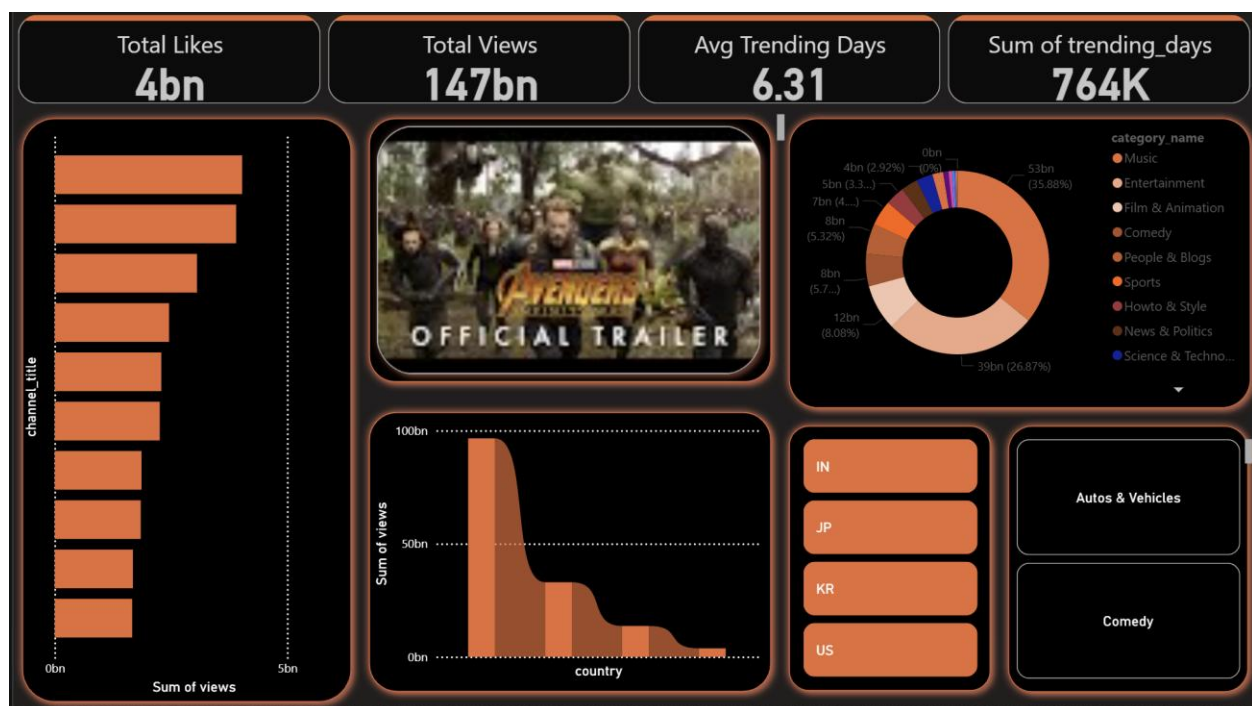
- **Google Colab** – For data processing and cleaning
 - **Python** – For data manipulation and feature engineering
 - **Pandas library** – For data analysis
 - **Power BI** – For creating the interactive dashboard
 - **Kaggle Dataset** – Source of real YouTube trending data
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Steps Involved in Building the Project:

- Downloaded the YouTube Trending dataset from Kaggle.
- Loaded the dataset into Google Colab for preprocessing.
- Cleaned the data by removing null values and duplicates.
- Merged JSON category files to get category names.

- Created new calculated columns such as:
 - Engagement Score
 - Like Ratio
 - Trending Days
- Exported the cleaned dataset as a CSV file.
- Imported the dataset into Power BI.
- Designed an interactive dashboard using charts, cards, slicers, and images.
- Analyzed trends and extracted insights from the dashboard.

Dashboard:



Conclusion:

This project provided hands-on experience in handling real-world datasets, data cleaning, and creating professional dashboards. It improved understanding of data visualization and storytelling through analytics. The project successfully demonstrated how trends and user engagement can be analyzed using modern business intelligence tools.